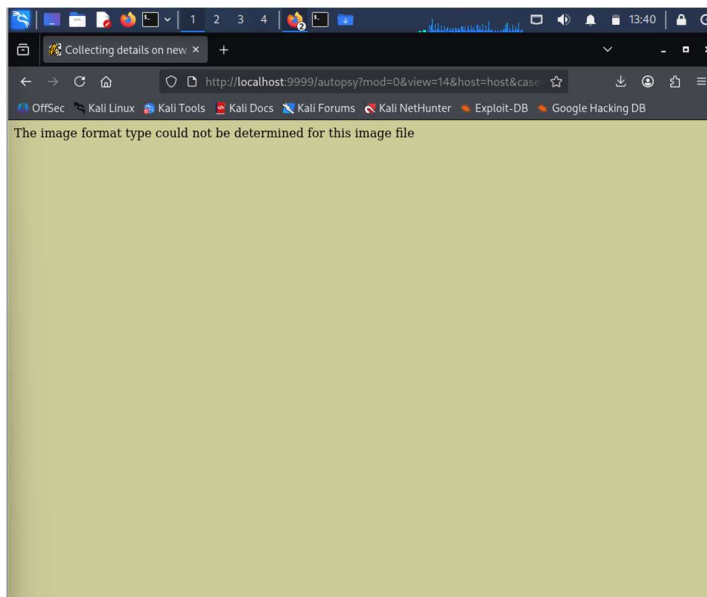


Insider

Objective: Analyzing Linux disk image artifacts, logs and bash history, using FTK Imager to investigate insider threat activities and reconstruct user actions

Unzipped then decided to use Autopsy in my Kali Linux instead of FTK Imager because Autopsy was already there.

Sudo autopsy



Not working.... Because it is an .ad1 file containing Linux File System.

Then realized that this is a 'LINUX' file system: meaning that I could just work through them using the terminal. No FTK Imager or Autopsy necessary.

Then realized I do indeed need it... I forgot to go all the way into the FirstHack.ad1 file, I had imported just the folder that the ad1 file was located in.

BUT AUTOPSY 2 CAN'T READ .ad1 files!!

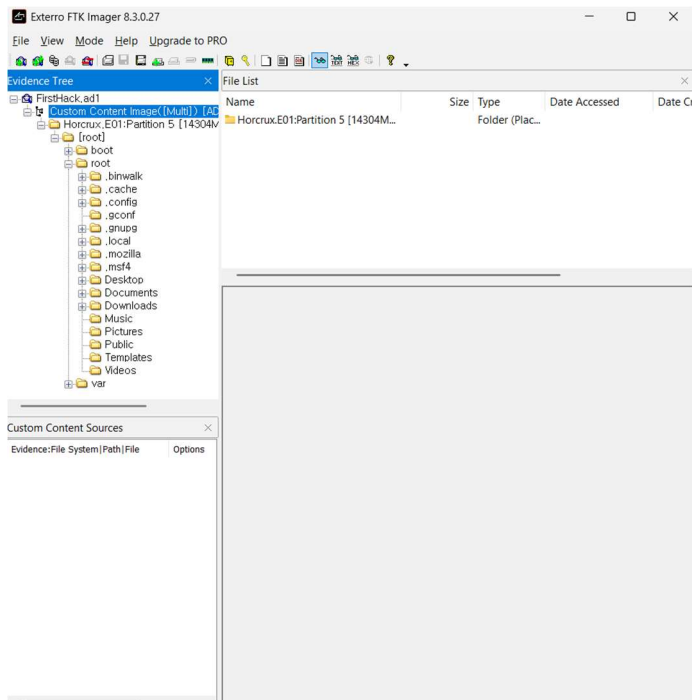
Eventually I came to the understanding on why the lab wanted me to use FTK Imager... no more kali linux, let's just use my Windows (Exits out of kali).

Tried downloading FTK Imager through its website but then faced another problem: I

have to register with a 'business email' which I don't have.

➔ Decided to just use a random business email as there are no verification process

All that just to get things set up... and they told me this CTF takes 45 minutes.



Finally.

Which Linux distribution is being used on this machine?

Went through different directories and being honest, all the files in the boot directory had the word kali so I figured it was kali.

What is the MD5 hash of the Apache access.log file?

➔ Decided to look through /var/log

But what is a MD5 Hash?

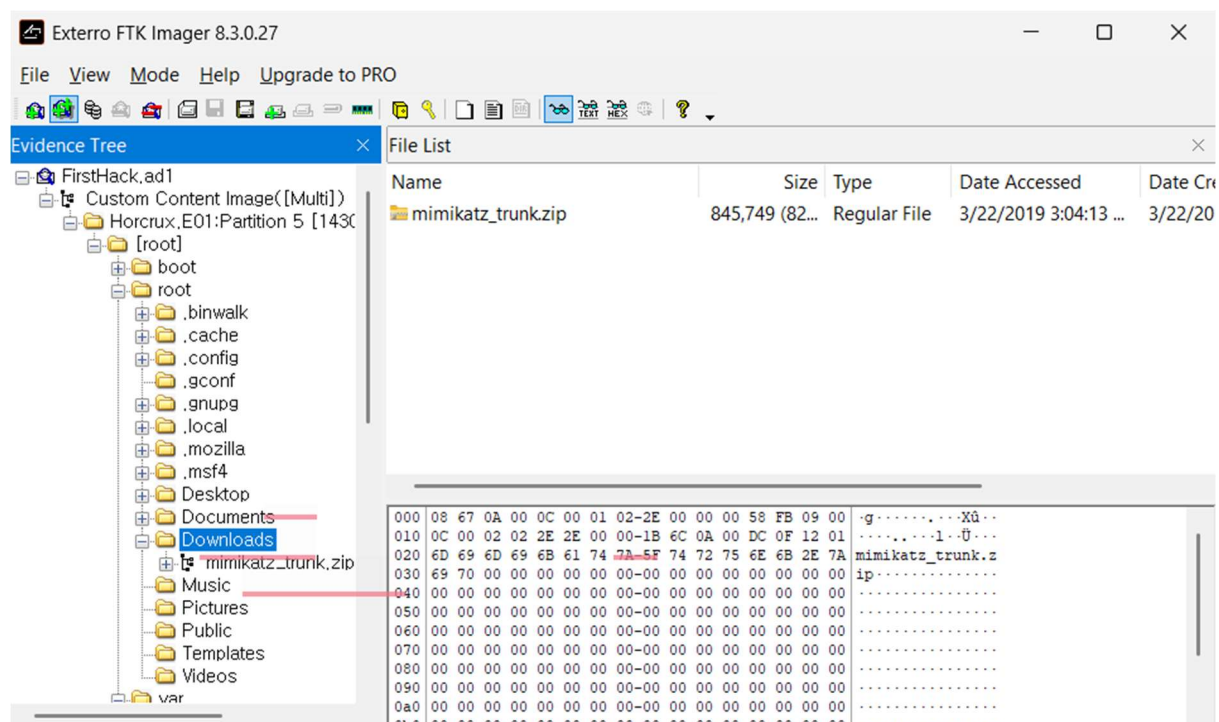
- Unique digital fingerprint for a file (32 characters) – basically MAC address for a file's inside contents, not the file itself.
- Right clicked the access.log file and exported hash file:

MD5,SHA1,FileNames

"d41d8cd98f00b204e9800998ecf8427e", "da39a3ee5e6b4b0d3255bfef95601890afd80709", "FirstHack.ad1\Custom Content Image([Multi]) [AD1]\Horcrux.E01:Partition 5 [14304MB]:NONAME [ext4]\[root]\var\log\apache2\access.log"

It is suspected that a credential dumping tool was downloaded. What is the name of the downloaded file?

➔ Going through the Downloads folder

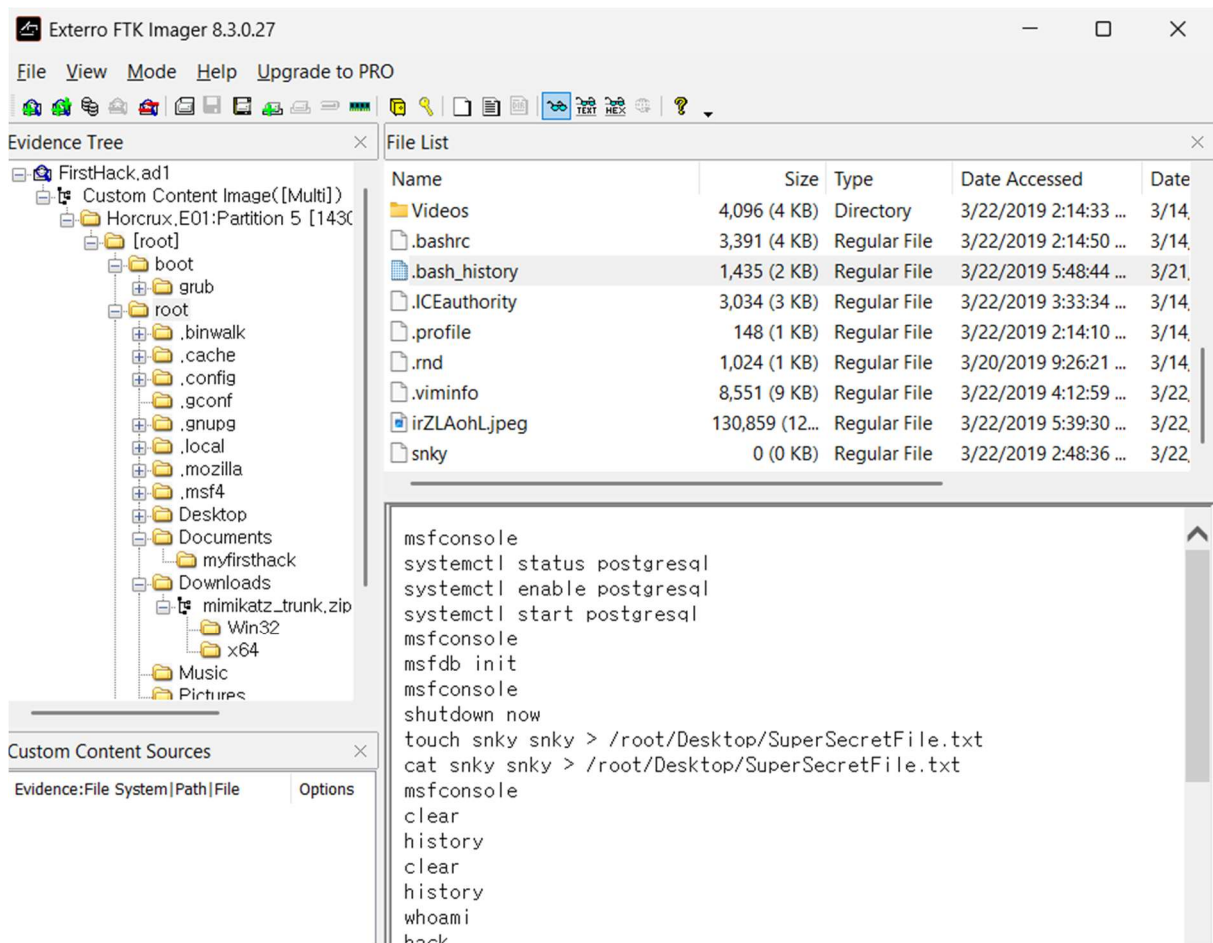


Well it was right there = mimikatz_trunk.zip

A super-secret file was created. What is the absolute path to this file?

After some research, I come to understand the importance of .bash_history file

➔ Hidden text file that stores a list of all the terminal commands typed in the bash shell



➔ /root/Desktop/SuperSecretFile.txt

What program used the file *didyouthinkwedmakeiteasy.jpg* during its execution?

➔ binwalk didyouthinkwedmakeiteasy.jpg was one of the commands in .bash_history – meaning that **binwalk** used that file.

What is the third goal from the checklist Karen created?

Found the checklist in the Desktop folder – people usually put those things in Desktop I guess

Check List:

- Gain Bob's Trust

- Learn how to hack

- Profit

How many times was Apache run?

What does it mean for Apache to run # times? What exactly is Apache?

Apache: web server software, bridge between a website's files and people visiting the site(delivery)

Parent process: master controller based on config settings

Child processes: workers created by parent to handle traffic

Parent + child = # of apache ran

APACHE == BASICALLY OS FOR WEBSITES

- ➔ /var/logs/apache2/ shows evidence of Apache activity
- ➔ Files were all empty
- ➔ Means Apache was not used

== Matches with the MD5 hash of access.log(d41d8cd98f00b204e9800998ecf8427e) because that hash is famously known as an empty file has(zero bytes).

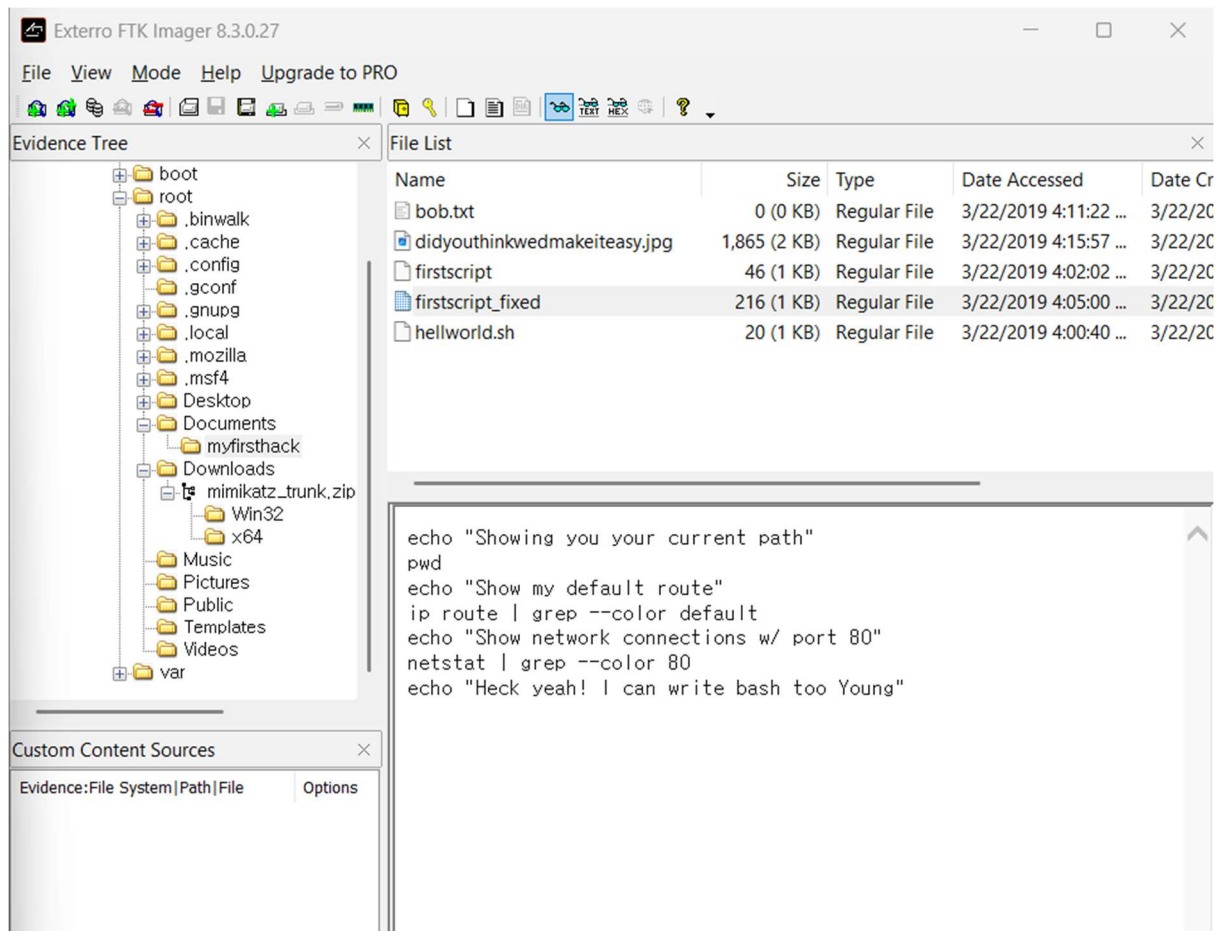
This machine was used to launch an attack on another. Which file contains the evidence for this?

Found a weird image jpeg file in the Desktop folder, which shows a command prompt that has Bob as the user..... (irZLAohL.jpeg)

- ➔ This, along with the checklist I found earlier, might suggest Karen is using this machine to launch an attack on Bob.

It is believed that Karen was taunting a fellow computer expert through a bash script within the Documents directory. Who was the expert that Karen was taunting?

- ➔ So I obviously looked through the Documents directory



→ **Young** is a crazy name

A user executed the su command to gain root access multiple times at 11:26. Who was the user?

- Tried looking through the .bash_history file again
- No time listed nor were there su commands
- Gotta go hint
- Found that /var/log/auth.log was the system's authentication logs
- Auth.log file contains records of all su commands

What is a su command(I don't know this)

Su (substitute/switch user): linux command line tool to temporarily log into another user account without logging out of your current session

Su: becoming another user(root) until you exit: needs root password

Sudo: borrowing another user(root) for one command: needs your own password and permission from root/admin

```

Mar 20 11:26:22 KarenHacker su[4060]: Successful su for postgres by r
Mar 20 11:26:22 KarenHacker su[4060]: + ??? root:postgres
Mar 20 11:26:22 KarenHacker su[4060]: pam_unix(su:session): session c
Mar 20 11:26:22 KarenHacker su[4060]: pam_systemd(su:session): Cannot
Mar 20 11:26:22 KarenHacker su[4074]: Successful su for postgres by r
Mar 20 11:26:22 KarenHacker su[4074]: + ??? root:postgres
Mar 20 11:26:22 KarenHacker su[4074]: pam_unix(su:session): session c
Mar 20 11:26:22 KarenHacker su[4074]: pam_systemd(su:session): Cannot
Mar 20 11:26:22 KarenHacker su[4081]: Successful su for postgres by r
Mar 20 11:26:22 KarenHacker su[4081]: + /dev/pts/0 root:postgres
Mar 20 11:26:22 KarenHacker su[4081]: pam_unix(su:session): session c
Mar 20 11:26:22 KarenHacker su[4081]: pam_systemd(su:session): Cannot
Mar 20 11:26:22 KarenHacker su[4094]: Successful su for postgres by r
Mar 20 11:26:22 KarenHacker su[4094]: + /dev/pts/0 root:postgres
Mar 20 11:26:22 KarenHacker su[4094]: pam_unix(su:session): session c
Mar 20 11:26:22 KarenHacker su[4094]: pam_systemd(su:session): Cannot
Mar 20 11:26:22 KarenHacker su[4101]: Successful su for postgres by r
Mar 20 11:26:22 KarenHacker su[4101]: + /dev/pts/0 root:postgres
Mar 20 11:26:22 KarenHacker su[4101]: pam_unix(su:session): session c
Mar 20 11:26:22 KarenHacker su[4101]: pam_systemd(su:session): Cannot
Mar 20 11:26:23 KarenHacker su[4101]: pam_unix(su:session): session c
Mar 20 11:26:23 KarenHacker su[4114]: Successful su for postgres by r
→ Mar 20 11:26:23 KarenHacker su[4114]: + /dev/pts/0 root:postgres

```

Postgres is the user, KarenHacker is just the machine hostname assigned.

Based on the bash history, what is the current working directory?

- Back to the .bash_history file
- Runs touch keys.txt at the second-to-last entry
- This probably means the current working directory is the directory where keys.txt is located
- Looked for keys.txt
- Too much work...
- Realized I could just look for the latest cd command
 - cd ../Documents/myfirsthack/
- So the directory was: **/root/Documents/myfirsthack/**

So What did Karen Actually do here? (recon) :

The Checklist above showed her motives (Gain bob's trust, learn hacking, and profit)

She also did not like Young I guess.

Kali Linux was running on Karen's machine, with *mimikatz_trunk.zip* downloaded, which turns out to be a tool use to steal plaintext passwords and hash credentials(encoded irreversible passwords) from computer memory.

SuperSecretFile.txt: used to store extracted hidden data/payloads from *didyouthinkwedmakeiteasy.jpg*(very creative) using tool called *binwalk*

Enter through Postgres(database) user session and su command to get root privileges

Network attack on Bob's computer

Left *irZLAohL.jpeg* accidentally(or purposefully I don't know) proof of her work

Last active location was in the */root/Documents/myfirsthack/* directory, and *touch keys.txt* was one of the last commands.

➔ Meaning this file may have been created to store her findings.

Takeaways:

- *Sometimes the setup is more complicated and getting the tools(FTK) set up takes longer.*
- *Once again, research is an essential part of this work: not gonna memorize every single path, every single tool, or command syntax*
- *Knowing what action to take is more important*
- *More experience = Higher adaptability = more familiarity with tools or commands to use*
- *Fun and enjoyable work once again; it feels like being a cyber detective reconstructing scenarios through digital evidence! Hope to learn more and be better.*